CONCORD v0.3 Runtime Contract Schemas

Normative runtime schema companion for the CONCORD specification pack

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Purpose: This document defines the runtime contract schemas omitted from the initial CONCORD v0.3 pack. It standardizes action contracts, state contracts, budget entities, coordination tokens, telemetry outputs, and a worked reference resource so implementations do not drift by local convention.

1. Scope and relationship to the spec pack

This document complements the Core Specification, Scan Output Schemas, and Error and Conflict Code Catalog. It standardizes the data contracts that implementers use for day-to-day runtime behavior.

- ActionContract is the atomic execution unit exposed to agents and control services.
- StateContract is the workflow declaration format for business state and guarded transitions.
- BudgetProfile and BudgetLedger define bounded execution policy.
- Lease and Token define portable coordination primitives.
- CoordinationTelemetry defines the portable runtime metrics surface.
- The worked example shows how the contracts fit together for one reference resource type.

2. Normative conventions

Unless explicitly marked optional, fields in the following schemas are required. Defaults apply only when the producer omits the field and the schema allows omission.

Term	Meaning	Implementation effect
Required	Field or rule that MUST be implemented	Omission is non-conformant
Optional	Field or rule that MAY be implemented	Safe to omit unless profile elevates it
Default	Normative value applied when omitted	Consumers may assume this value
Profile-specific	Requirement activated by deployment profile	Minimum profile may defer it

3. ActionContract schema

ActionContract defines a named mutating or non-mutating capability. It is the canonical contract consumed by admission control, policy, budgeting, and agent planning.

3.1 Core fields

Field	Type	Req	Rule / default
action_name	string	Y	Stable unique action identifier
description	string	Y	Human-readable summary
resource_type	string	Y	Target resource class
action_family	enum	Y	read, plan, mutate, approve, deploy, compensate, admin
input_schema_ref	string	Y	Reference to input schema artifact
output_schema_ref	string	Y	Reference to output schema artifact
guard_predicates	array<object>	Y	One or more named preconditions
required_capabilities	array<string>	Y	CapabilitySet grants needed to admit
required_trust_tier	enum	N	Default T1/propose for non-read, T0/observe for read
idempotency_required	boolean	Y	True for all mutating actions
idempotency_scope	enum	N	session by default; may be resource or global
retry_class	enum	N	none, safe_retry, recheck_then_retry, queue_then_retry
risk_level	enum	Y	low, moderate, high, critical

3.2 Coordination, recovery, and consistency fields

Field	Type	Req	Rule / default
side_effects	array<object>	N	Explicit downstream writes, events, or jobs
participates_in_saga	boolean	N	Default false
compensation_strategy	enum	N	none, inverse_action, saga_handler, manual_only
compensation_order_hint	integer	N	Lower number compensates first
consistency_profile	enum	Y	STRONG, SESSION_MONOTONIC, READ_YOUR_WRITES, EVENTUAL, ASYNC_PROJECTION
authoritative_recheck_required	boolean	N	Default true for EVENTUAL and ASYNC_PROJECTION mutation paths
projection_tolerance_ms	integer	N	Required when profile is ASYNC_PROJECTION
session_watermark_required	boolean	N	Default true for SESSION_MONOTONIC
conflict_resolution_strategy_override	string	N	Optional per-action override of resource default
budget_cost_units	number	N	Default 1.0
budget_dimensions_consumed	array<string>	N	Examples: mutate, high_risk, deploy

3.3 Normative rules

- Every mutating action MUST declare idempotency_required = true and a non-empty idempotency scope.
- Every action MUST declare exactly one consistency_profile.
- Actions that rely on projected reads for planning MUST still submit to authoritative admission at intent execution time.
- If participates_in_saga = true, compensation_strategy MUST be present even when set to manual_only.
- A read action MAY set required_trust_tier to T0/observe and idempotency_required = false.
- Budget cost units SHOULD reflect risk-weighted operational cost rather than request count alone.

3.4 Reference JSON shape

```
{
  "action_name": "submit_change_request",
  "resource_type": "change_request",
  "action_family": "mutate",
  "input_schema_ref": "schemas/change_request_submit_input.json",
  "output_schema_ref": "schemas/change_request_submit_output.json",
  "guard_predicates": [
    {"name": "state_is_draft"},
    {"name": "required_fields_complete"},
    {"name": "edit_lease_held"}
  ],
  "required_capabilities": ["change_request_modify"],
  "required_trust_tier": "T2/bounded",
  "idempotency_required": true,
  "idempotency_scope": "resource",
  "retry_class": "recheck_then_retry",
  "risk_level": "moderate",
  "participates_in_saga": true,
  "compensation_strategy": "inverse_action",
  "compensation_order_hint": 20,
  "consistency_profile": "EVENTUAL",
  "authoritative_recheck_required": true,
  "budget_cost_units": 3.0,
  "budget_dimensions_consumed": ["mutate"]
}
```

4. StateContract schema

StateContract defines business workflow state. Coordination state remains separate and MUST NOT be collapsed into the business state graph.

4.1 State declaration fields

Field	Type	Req	Rule / default
resource_type	string	Y	Owning resource class
initial_state	string	Y	Single starting state
states	array<object>	Y	One or more declared states
transitions	array<object>	Y	One or more legal transitions
terminal_states	array<string>	Y	Subset of declared states;

			may be empty
rollback_rules	array<object>	N	State or transition-specific rollback rules
entry_hooks	array<object>	N	Optional state-entry hook declarations
exit_hooks	array<object>	N	Optional state-exit hook declarations

4.2 State object and transition object

Object	Field	Req	Rule / default
state	name	Y	Stable state identifier
state	description	N	Human-readable meaning
state	is_terminal	N	Default false
state	allowed_action_refs	N	Action names typically available in this state
transition	name	Y	Stable transition identifier
transition	from_state	Y	Single origin state
transition	to_state	Y	Single destination state
transition	action_ref	Y	ActionContract that may trigger the move
transition	guards	Y	Named predicates evaluated before admission
transition	on_success	N	Receipts, events, or hook references
transition	rollback_to_state	N	Optional rollback target

4.3 Normative rules

- initial_state MUST appear in states.
- Each transition MUST reference a declared action_ref.
- No transition MAY target an undeclared state.
- Terminal states MAY still allow read or review actions, but they MUST NOT allow workflow-advancing mutation unless explicitly declared.
- Rollback rules MUST name either a rollback target state or a compensation handler.
- Business state declarations MUST NOT encode lease ownership, queue position, or budget state.

5. BudgetProfile and BudgetLedger schemas

Budget entities define bounded execution. Gateway checks are preliminary; intent-layer admission remains authoritative for action-specific budget decisions.

5.1 BudgetProfile

Field	Type	Req	Rule / default
budget_profile_id	string	Y	Stable unique identifier
max_actions_per_minute	integer	Y	Coarse request budget
max_mutating_actions_per_hour	integer	Y	Mutation budget
max_high_risk_actions_per_day	integer	Y	High-risk budget
max_cost_units_per_session	number	Y	Risk-weighted session budget
max_parallel_leases	integer	N	Default 1

max_queue_depth	integer	N	Optional queue slot ceiling
burst_limit	integer	Y	Gateway burst threshold
cooldown_policy	enum	Y	fixed_backoff, exponential_backoff, contention_scaled, risk_scaled, manual_resume_required
circuit_breaker_thresholds	object	Y	Threshold bundle for throttle/open actions
per_resource_class_limits	object	N	Optional class-specific ceilings
per_action_family_limits	object	N	Optional action family ceilings

5.2 BudgetLedger

Field	Type	Req	Rule / default
ledger_id	string	Y	Stable unique identifier
session_id	string	Y	Owning session
agent_class_id	string	Y	Owning agent class
budget_profile_id	string	Y	Applied budget profile
window_start	timestamp	Y	Observation window start
window_end	timestamp	Y	Observation window end
actions_consumed	integer	Y	Requests admitted
mutating_actions_consumed	integer	Y	Mutations admitted
high_risk_actions_consumed	integer	Y	High-risk actions admitted
cost_units_consumed	number	Y	Accumulated weighted cost
parallel_leases_in_use	integer	Y	Live lease count
queue_slots_in_use	integer	Y	Live queue reservations
circuit_state	enum	Y	closed, throttled, open
cooldown_until	timestamp	N	Required when throttled or open

6. Lease and Token schemas

Leases provide short-lived ownership. Tokens provide capacity or quorum style coordination where a resource or workflow path has bounded parallelism.

6.1 Lease

Field	Type	Req	Rule / default
lease_id	string	Y	Stable unique identifier
resource_id	string	Y	Target resource
lease_type	enum	Y	edit, review, execute, reservation, approval_slot
holder_session_id	string	Y	Owning session
purpose	string	N	Human or machine-readable rationale
granted_at	timestamp	Y	Grant time
expires_at	timestamp	Y	Lease must expire
renewal_count	integer	N	Default 0
status	enum	Y	active, expired, released, revoked

6.2 Token

Field	Type	Req	Rule / default
token_id	string	Y	Stable unique identifier
resource_id	string	Y	Owning resource or coordination path
token_type	enum	Y	capacity, quorum, validation_gate, deployment_gate
capacity	integer	Y	Total token count
available_count	integer	Y	Unallocated count
holders	array<string>	Y	Current holder session ids or role ids
issuance_rule	string	Y	Rule or policy reference for allocation
expires_at	timestamp	N	Optional expiration
status	enum	Y	active, exhausted, expired, suspended

7. CoordinationTelemetry schema

CoordinationTelemetry is the portable metrics surface emitted by a CONCORD runtime. It allows dashboards, scanners, and controllers to reason about contention and degradation consistently across implementations.

Field	Type	Req	Rule / default
telemetry_window_start	timestamp	Y	Observation window start
telemetry_window_end	timestamp	Y	Observation window end
resource_type	string	N	Optional resource filter
lease_contention_rate	number	Y	Contended lease requests / total lease requests
average_queue_wait_ms	number	Y	Mean wait time
queue_abandonment_rate	number	Y	Abandoned queue entries / total queue entries
stale_write_rejection_rate	number	Y	STALE_VERSION rejects / mutating attempts
budget_throttle_rate	number	Y	Budget throttles / admissions
circuit_breaker_trip_count	integer	Y	Trips in window
compensation_invocation_rate	number	Y	Compensations / saga starts
arbitration_frequency	number	N	Optional advanced coordination metric
resource_hotspots	array<object>	N	Ranked hotspots by contention or failure density
retry_distribution	object	Y	Histogram keyed by retry class

7.1 Hotspot entry object

Field	Type	Req	Rule / default
resource_id	string	Y	Hotspot resource
hotspot_score	number	Y	Composite contention/failure score
primary_cause	string	Y	Dominant contention or failure driver

recommended_action	string	N	Operational remediation hint

8. Worked reference example: change_request

The reference resource below is intentionally small but complete. It shows how one resource type should be specified across actions, workflow, budgeting, and advisory planning surfaces.

8.1 State contract summary

State	Terminal	Typical actions	Notes
draft	N	update_change_request, request_edit_lease, submit_change_request	Initial authoring state
submitted	N	request_review_token, withdraw_change_request	Awaiting reviewer action
under_review	N	approve_change_request, reject_change_request	Review and validation in progress
approved	Y	read_change_request, deploy_change_request	Workflow complete; deployment may be separate
rejected	Y	read_change_request, reopen_change_request	Terminal for workflow; reopen is explicit new path

8.2 Action contract summary

Action	Family	Consistency	Risk	Key guards
update_change_request	mutate	STRONG	low	state_is_draft, edit_lease_held
submit_change_request	mutate	EVENTUAL	moderate	required_fields_complete, authoritative_recheck
approve_change_request	approve	READ_YOUR_WRITES	high	state_is_under_review, review_token_held
deploy_change_request	deploy	ASYNC_PROJECTION	critical	approved, human_gate_satisfied, projection_within_tolerance

8.3 Budget profile summary

Field	Value	Rationale
max_actions_per_minute	60	Allows advisory reads and moderate control traffic
max_mutating_actions_per_hour	24	Prevents mutation storms on a single session
max_high_risk_actions_per_day	3	Critical actions remain tightly bounded
max_cost_units_per_session	40	Risk-weighted cap across the workflow
cooldown_policy	contention_scaled	Backoff rises under hotspot conditions

8.4 Sample OperationContext response

```
{
  "resource_id": "cr_2048",
  "resource_type": "change_request",
  "business_state": "submitted",
  "coordination_state": "review_token_available",
  "version": 12,
  "allowed_actions": ["request_review_token", "withdraw_change_request"],
  "blocked_actions": ["approve_change_request"],
  "blocked_reasons": ["state_not_under_review"],
  "active_leases": [],
  "queue_position": null,
  "pending_intents": ["intent_9d3f"],
  "consistency_profile": "EVENTUAL",
  "projection_lag_ms": 120,
  "budget_remaining": {"cost_units": 31.0, "high_risk_actions_today": 3},
  "trust_constraints": ["deploy_change_request requires T4/governed_critical"],
  "recommended_next_action": "request_review_token",
  "safe_parallel_actions": ["read_change_request"],
  "requires_authoritative_recheck": true,
  "context_assembled_at": "2026-03-03T10:15:22Z",
  "context_ttl_ms": 5000,
  "freshness_status": "fresh"
}
```

9. Minimum implementation profile and deferred elements

A minimum conformant runtime MAY defer advanced coordination features, but it MUST implement the core runtime contracts listed below.

Included in minimum runtime	May be deferred
ActionContract, StateContract, Session, Intent, Resource, basic exclusive Lease, OperationContext, Receipt, BudgetProfile, BudgetLedger	Queues, fairness policies, negotiation protocols, advanced quorum tokens, arbitration services
Version checks, idempotency, consistency profiles, gateway and intent-layer budget enforcement	Advanced conflict strategy variants beyond default
Saga timeout and poison handling when sagas are used	Complex multi-party external_gated orchestration beyond declared contracts

10. Implementation notes

- Schema producers SHOULD publish machine-readable JSON Schema or equivalent from these normative definitions.
- Runtime responses SHOULD include schema version identifiers so clients can negotiate additive changes safely.
- If a deployment omits Token, it SHOULD still reserve the type name for future compatibility.
- CoordinationTelemetry SHOULD be exportable both as point-in-time JSON and as time-series friendly records.
- Worked examples SHOULD be shipped with implementation test fixtures so scanners and agents can validate behavior against known-good contracts.